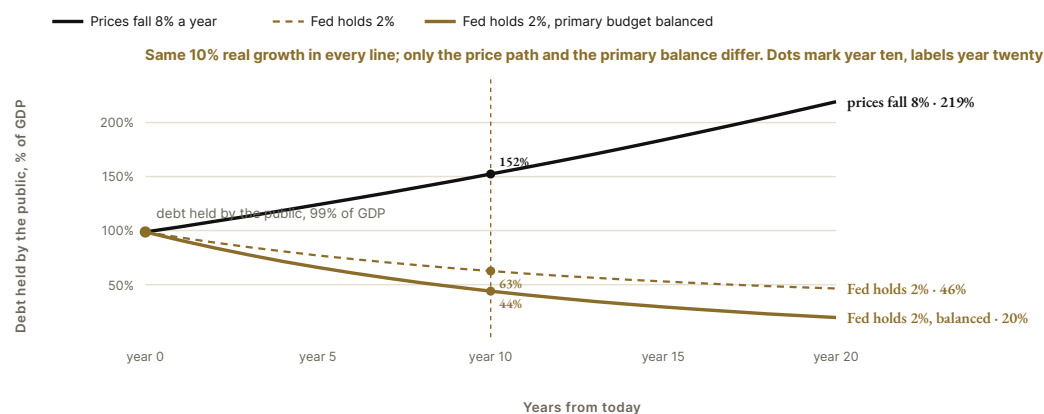


THE ABUNDANCE LEDGER · PART 2 OF 2 · FISCAL & MACRO

Robots, AI, Abundance, and the Debt

If robots and AI deliver the abundance their proponents describe, what stays scarce, does a basic income get funded, does the national debt get paid down, and what did the one moneyless economy in science fiction have to build first? Four questions, in the order the argument runs, on the industry's own numbers.

TON618 CAPITAL RESEARCH · TREASURY FISCAL DATA THROUGH 3 SEPTEMBER 2026 · BEA, BLS AND CBO THROUGH THE 2026 Q2 ACCOUNTS AND THE 2026 OUTLOOK · PART 1 OUTPUTS AS OF 5 SEPTEMBER 2026



How to read: three of the debt paths from question three. The horizontal axis counts years from today, to twenty; the vertical axis is federal debt held by the public as a share of GDP, starting from today's 99%. Dots mark year ten, the end labels year twenty. Every line has the same robots, the same 10% real growth and the same interest rate on the debt. The only things that differ are whether prices are allowed to fall and whether the primary budget is balanced.

Four questions, in the order the argument runs

Four questions, asked in the order the argument runs, and four answers on the industry's own numbers. The abundance case says robots and AI make everything cheap, cheap everything makes a citizen's income go further, growth pays the debt, and a basic income covers whoever is displaced. Each link is a question with a number behind it. The ledger runs ten to twenty years, the horizon Musk himself gives for work becoming optional ("maybe it's ten, twenty years, something like that"; for money he gives none, only "at some point in the future"), with the arithmetic reported at ten years and at twenty. Where the note reasons past that horizon, to the limit the proponents describe, it says so and confines the reasoning to one section, labeled as a thought experiment.

Does the robot make things cheap? Yes, for the part of the economy a factory makes. On the industry's forecast, robot labor reaches 1% of US hours worked in 2028 and 58% in 2035; on Part 1's consensus path, 1% around 2032 and about 5% by 2035. The price of a fixed unit of intelligence fell 78 to 114 times in under two years, the strongest fact in the proponents' favor in either note, and goods excluding food and energy are up only 16% since 2000 while durable goods are down 2%.

What stays expensive? Everything with a fixed factor behind it. Rent is up 148% since 2000, electricity 141% and medical care 133%, and shelter, energy and medical care are half the consumer price index. Land's share of the value of a home rose from 36% to 59% between 2012 and 2024. The surplus from the one input the robot has cheapened, labor, is flowing to the owners of the inputs it has not.

What do falling prices do to the debt? They decide the case. The same 10% real growth either halves the federal debt ratio in nine years if the primary budget also closes, or raises it by half, depending on one variable that has nothing to do with robots: whether prices are allowed to fall. With prices falling 8% a year, debt held by the public goes from 99% of GDP to 152% in a decade and 219% in two. With the Federal Reserve holding 2% and the primary budget balanced, it goes to 44% and then 20%. Those are fixed-coupon numbers; let the coupon reprice toward growth and the two paths converge, 136% and 155% at twenty years, and the primary deficit decides. Japan ran the deflation case from 1998 to 2012 and its debt went from 102% of GDP to 197%. The abundance argument is a bet

on Fed policy, which its academic form states openly and its popular form does not. And a large part of what the government owes is a promise of care rather than of dollars: Medicare and the health programs, 29% of outlays so far this fiscal year, are inflated away in no row and deflated away in none; they fall only if the cost of a surgery falls, half of which is labor, which is where the proponents' case is strongest.

Who pays, and can a basic income be funded? Capital, and only if a claim on it is legislated first. Individual income and payroll taxes are 87% of federal receipts this fiscal year. Replacing them from capital income alone takes a claim equal to 29% of all non-labor income; a poverty-line basic income at its net cost takes it to 33%, and at its gross cost to 56%. Growth does not change the 29%; it shrinks the basic-income increment, to about 0.7% of GDP for a poverty floor after a decade at 10% growth, without changing where either is collected from. Ownership does. Every serious model, from Korinek and Lockwood's public-finance primer to Altman's equity fund, arrives at a claim on capital or land, and none of them has a statute.

Why Star Trek. The claim this note tests is literal, and it is the proponents' own. Elon Musk, at the U.S.-Saudi Investment Forum in November 2025: "My prediction is that work will be optional," and, "if you go out long enough, assuming there's a continued improvement in AI and robotics, which seems likely, money will stop being relevant at some point in the future." At VivaTech in 2024 he called the end state "universal high income" with "no shortage of goods or services." He draws the picture from fiction and says so, recommending Iain Banks's Culture novels, in which "money no longer exists." A claim that money stops being relevant is a claim about what an economy without money would need, and it is not spelled out there. So the note borrows the one moneyless economy most readers already know, Star Trek's Federation, and asks what its writers had to give it before money could be dropped. The answer is three machines, none of them a robot, standing for energy, matter and land rather than labor; questions one and two measure what has happened to the price of the inputs those machines stand for, and the closing section returns to the franchise's own account of what happened when its economy lost one.

Fact versus opinion, stated up front. Every debt, receipt, price and income figure here is a filed or official statistic; the data-center electricity figures are the International Energy Agency's, from its 2025 report, and the surgery counts are the orthopaedic registry's. The robot-hour path is the industry's forecast and is

labeled as such throughout. The house takes no position on whether the Fed should target nominal income, whether a basic income is good policy, or whether land should be taxed. This note shows what the arithmetic requires under each choice and what would change the reading.

⊕ THE OTHER SIDE FIRST

The abundance case in its strongest form

The strongest version of the abundance argument is not "robots make everything free." It is four specific claims, each with a serious literature behind it.

1. **Growth pays debts, and the growth on offer is unprecedented.** Brookings' 2026 simulation of a conventional AI productivity shock cuts the federal deficit by about five percentage points of GDP by 2036, from roughly 6% to roughly 2%, through the tax base alone. The Yale Budget Lab, running the median economist's 2.5% productivity forecast through its model, finds the debt path more sustainable even after labor-force participation falls and displaced-worker outlays rise; only its most generous transfer variant offsets the gain. Penn Wharton scores \$400 billion of deficit reduction over ten years on a far smaller productivity effect. A 10% real growth rate would be four times the largest of those shocks, and the arithmetic of a debt ratio is that growth in the denominator does most of the work.
2. **The fixed factors are not fixed.** Land is scarce only where zoning makes it so, and a country with 1.9 billion acres is not short of it. Energy is a cost curve, not a constraint: Part 1's reference class puts solar modules and battery cells at a 20% learning rate per doubling, and Musk's own version of the argument is energy-first, that cheap electrons come before cheap labor. Health care is half labor, and labor is the input AI is replacing. Every one of the three bottlenecks this note leans on has a technology aimed at it.
3. **The tax base problem is known and solvable by design.** Henry George wrote the answer in 1879: abolish all taxation save that upon land values. Altman's 2021 proposal is a 2.5% annual levy on large-company equity and on privately held land, paid into a fund that distributes to every adult. Korinek and Lockwood's 2026 primer maps the whole sequence, from consumption taxes in a post-labor economy to a harvesting tax on

autonomous capital, and finds that sovereign wealth funds and public equity stakes capture AI returns without distorting investment. A comprehensive system of fixed-factor taxation, they estimate, could raise 5% to 10% of GDP. The Nayebi threshold model finds that raising public capture of AI rents from today's 15% to a third halves the capability the machines need to fund a basic income.

4. **The price path is a policy choice, and the policy exists.** Deflation is not a law of productivity. A central bank that targets the level of nominal income rather than a 2% inflation rate lets productivity gains arrive as nominal growth instead of falling prices, which is exactly the case in which the debt ratio falls fastest. The proposal has a literature, a name and advocates inside the Fed system. Every debt scenario in this note that goes wrong goes wrong because prices fall; the proponents' answer is not to let them.
5. **What this note cannot test within its frame.** It cannot test whether the robot hours arrive; it takes Part 1's industry path as given. It cannot test future Fed doctrine or future tax law, and it cannot test whether the Treasury's coupon would follow prices down in a deflation, which the Japanese record suggests it would. Of the four arguments above, the arithmetic tests 1 directly, tests 2 against the price record, and tests 3 by computing the size of the claim the design would have to make; it takes 4 as the decisive variable and shows what turns on it rather than predicting it.

The four questions that follow must beat this case, not a weaker one.

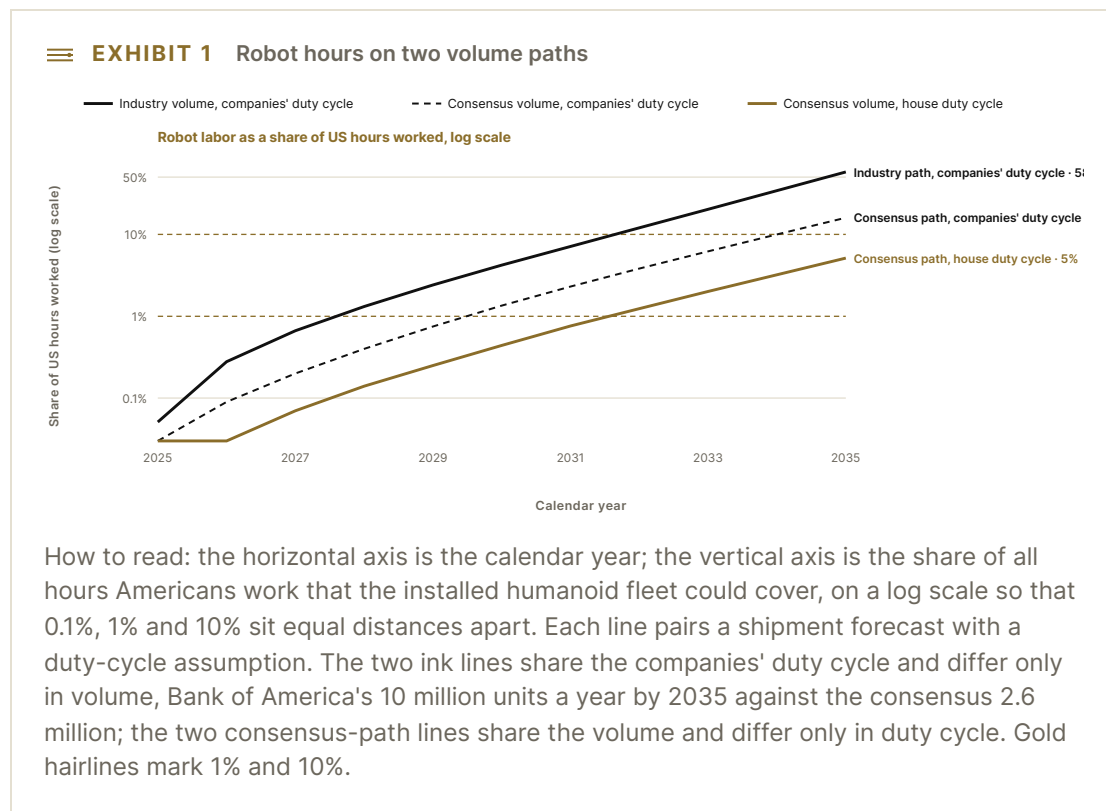
↪ QUESTION ONE

Does the robot make things cheap?

The abundance argument begins with hours, because a robot makes things cheap by doing for the cost of its depreciation, its maintenance and whatever supervision it still needs what a worker does for a wage. Part 1 measured what a robot-hour costs and found that price was not the binding variable; this note needs the other side of that ledger, how many robot-hours the industry's own forecasts imply against the hours Americans work, because everything downstream, the price level, the debt and the tax base, scales with that number.

The United States works about 285 billion hours a year, 159 million payroll jobs at 34.4 hours a week. Bank of America forecasts humanoid shipments of 20,000 in 2025, 90,000 in 2026 and 10 million a year by 2035, an 86% compound growth rate. Run at the duty cycles the companies themselves assume in Part 1, with robots retired after five years, the share of US hours worked that the installed fleet could cover is:

YEAR	ROBOTS INSTALLED (5- YEAR LIFE)	COMPANIES' CASE: 20 H/DAY, 99% SUCCESS	TWO SHIFTS, 85% SUCCESS	ONE SHIFT, 85% SUCCESS, 3-YEAR LIFE
2028	518,000	1.3%	0.9%	0.4%
2030	1.66 million	4.2%	2.9%	1.2%
2032	4.73 million	12.0%	8.3%	3.5%
2035	22.7 million	57.8%	39.7%	17.0%



On Part 1's consensus volume path, 2.6 million units a year by 2035 rather than 10 million, the same duty cycles give 16% of US hours in 2035 at the companies' assumptions and 5% at the house's, with 1% arriving around 2032.

Two things about this table. First, it is the industry's forecast treated as if every robot shipped in the world were put to work in the United States, so it is an upper bound on the US path by construction; Part 1's house path, roughly a quarter of the volume, puts the first-percent milestone four years later and, extrapolating past the modeled window, the later ones about five. Second, even the conservative column on the industry path, one shift and a three-year life, reaches a sixth of US hours inside the CBO's current budget window. The fiscal arithmetic in questions three and four is therefore a question for this decade if the industry is right and for the next if the consensus is, and the note runs it on the industry's path because that is the case the abundance argument makes.

To replace 10% of US hours at the companies' duty cycle takes 3.9 million working humanoids; to replace all of them takes 39 million. For scale, there are 4.7 million industrial robots in operation worldwide.

What the hours have already bought: intelligence. Part 1 found that the price of a fixed unit of intelligence fell 78 to 114 times in under two years and then went flat, and that frontier deflation continues at four to five times a year. A fixed unit of intelligence at one seventy-eighth of its early-2023 price is the strongest fact in the proponents' favor in either note. What it has not touched is the price of anything with land, fuel or a licensed human in it, and that is question two.

What the hours have bought at the checkout. The goods that factories make have followed the same direction, more slowly. Since January 2000, commodities excluding food and energy are up 16% and durable goods are down 2%, against 97% for the index as a whole. Manufactured goods have been deflating in relative terms for twenty-five years, before a single humanoid was shipped; the robot extends a curve that already exists.

The answer to question one. Yes, for what a factory or a data center makes, and on the industry's path the hours arrive inside the budget window. The abundance argument is right about the part of the economy it looks at. The next question is the part it does not.

≡ QUESTION TWO

What stays expensive?

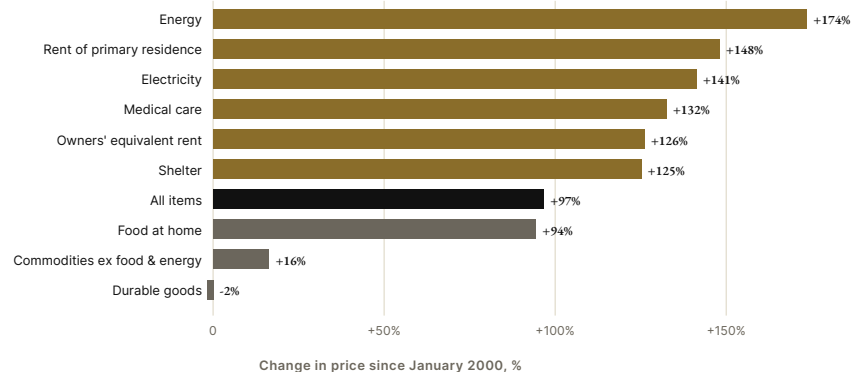
A citizen's income is spent on more than goods. The second question is what happens to the price of the rest: the ground under a home, the energy that runs

everything, and the care a licensed human provides. Labor is only one input into each of them, and the price record since 2000 shows what has happened to the others.

CONSUMER PRICE COMPONENT	CHANGE SINCE JANUARY 2000	WEIGHT IN THE INDEX, DECEMBER 2025
Energy	+174%	6.4%
Rent of primary residence	+148%	7.8%
Electricity	+141%	2.5%
Medical care	+133%	8.4%
Owners' equivalent rent	+126%	26.2%
Shelter, all	+125%	35.6%
All items	+97%	100%
Food at home	+94%	
Commodities excluding food and energy	+16%	19.2%
Durable goods	-2%	

EXHIBIT 2 Goods deflated; the fixed factors did not

Change in consumer prices since January 2000, %



How to read: each bar is one component of the consumer price index, and its length is how much that component's price has changed since January 2000, so a bar reaching +100% means the price doubled. Gold bars are the components with a fixed factor behind them, land, fuel or licensed labor: shelter, rent, owners' equivalent rent, energy, electricity and medical care. Gray bars are goods and food. The ink bar is the whole index.

What the record says. Goods deflated. Everything with a fixed factor behind it did the opposite. The basket of shelter, energy and medical care is half the consumer price index, and it has risen 132% since 2000 against 16% for the goods that factories make. This is the pattern Aghion, Jones and Jones formalized in 2017: when tasks are complements rather than substitutes, growth is constrained "not by what we are good at but rather by what is essential and yet hard to improve," and the capital share of income asymptotes near a third "even with nearly complete automation." The essential, hard-to-improve tasks set the price of living.

Land. Ricardo's 1817 result is the mechanism: rent is paid because land is "not unlimited in quantity and uniform in quality," and as demand rises the fixed factor absorbs the surplus. The American Enterprise Institute's land-price series shows it in real time. Land's share of the value of a US home rose from 35.7% in 2012 to 58.7% in 2024, and the value of an acre of residential land went from \$386,000 to \$1.49 million. Household real estate is \$53 trillion of a \$183 trillion household net worth. A robot can frame a house in an hour; it cannot make more of the places people want to live, and zoning fixes the supply of the places that exist. A basic income paid into that market becomes, at the margin, a transfer to whoever owns the land under it. The proponents' answer, that land is scarce only by regulation, is correct and is also an argument that the bottleneck is political rather than technical, which is not the same as saying it is not there.

Energy, and the machines are hungry. A human runs on about 100 watts. The humanoid in Part 1 draws 500 watts walking, and the data center behind it draws far more: the International Energy Agency puts data-center electricity at 415 terawatt-hours in 2024, 1.5% of world consumption, rising to about 945 by 2030, with the United States at 45% of the total and data centers accounting for nearly half of US demand growth over the period. Retail electricity has risen 141% since 2000, and the June 2026 rate is up about 5% on a year earlier. The energy cost curve the proponents cite is real at the module and the cell: Part 1's archive has the solar module at \$0.27 a watt in 2024 against about \$100 in 1976, a 99.7% decline. The price a household pays for a kilowatt-hour has not followed it. The gap between the module and the meter is transmission, distribution, capacity and permitting. The wires and transformers are manufactured goods with learning rates of their own, but the note does not model the grid's cost curve; it takes the 2000-to-2026 meter record as the evidence that on this window the curve has not reached the retail price, with data-center demand doubling against it, so the price at the meter does not follow the module on this

ledger's horizon; what happens past that horizon is the thought experiment before the closing section. The shape is the same as the gap between the token and the robot-hour in Part 1. In a robot economy where labor is free, the price of everything converges toward its energy content, and energy is the one input whose consumer price is rising faster than the index.

Medical care: the soft bottleneck. Health spending was 13.3% of GDP in 2000, 18.0% in 2024, and is projected at 20.6% by 2034. Half of it is labor that AI can plausibly reach. The other half is licensing, liability and reimbursement, which deflate at the speed of Congress.

The answer to question two. Of the three things a citizen's income must buy beyond goods, land, energy and care, none has deflated, and all three have risen faster than the index. The price record says the surplus from the one input the robot has cheapened, labor, is flowing to the owners of the inputs it has not.

⊖ QUESTION THREE

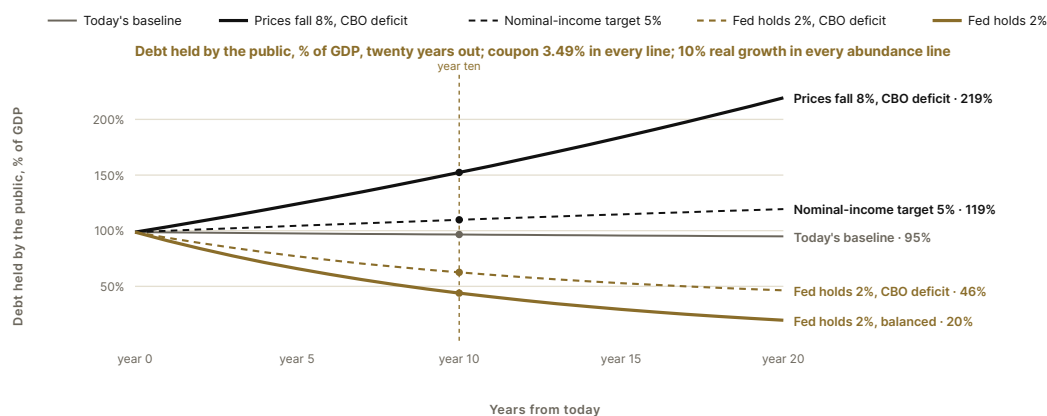
What do falling prices do to the debt?

The first two questions say the robot cheapens goods and intelligence and leaves the fixed factors alone. The proponents' claim is broader, that abundance lowers the price level itself, and the third question takes that claim as given and asks what it does to the debt. A government's debt ratio moves by a simple rule. Each year it grows by the interest rate on the debt, shrinks by the growth rate of nominal GDP, and moves by the primary balance, which is the deficit before interest. Nominal GDP is real output times the price level. That last word is where the abundance argument lives or dies, because a robot economy can deliver its productivity as more output at stable prices or as the same nominal output at falling prices, and the debt is a claim on dollars, not on output.

The inputs are today's. Debt held by the public was 98.7% of GDP in the March 2026 quarter; the CBO puts fiscal 2026 at 101%. The average interest rate on the government's interest-bearing debt was 3.49% in August 2026, against 3.4% in the CBO's projection. The primary deficit is 2.6% of GDP in the CBO's baseline. Recent nominal growth has been 6.6% a year with real growth of 2.1%. The table holds the coupon and the primary deficit fixed, sets real growth at the 10% the abundance argument implies, and varies only the price path.

SCENARIO, ALL AT 10% REAL GROWTH EXCEPT THE BASELINE	NOMINAL GROWTH	RATE MINUS GROWTH	DEBT/GDP, YEAR 5	YEAR 10	YEAR 20
Today's baseline: 2.1% real, recent inflation, CBO primary deficit of 2.6%	6.6%	-3.1 pts	98%	97%	95%
Abundance, prices falling 8%, CBO primary deficit	1.2%	+2.3 pts	124%	152%	219%
Abundance, Fed holds 2%, CBO primary deficit	12.2%	-8.7 pts	77%	63%	46%
Abundance, Fed holds 2%, primary budget balanced	12.2%	-8.7 pts	66%	44%	20%
Abundance, prices falling 8%, primary budget balanced	1.2%	+2.3 pts	110%	124%	154%
Abundance, Fed holds 2%, primary deficit of 8% (basic-income-sized)	12.2%	-8.7 pts	100%	101%	102%
Abundance, nominal-income level target at 5% (prices falling 4.5%), CBO primary deficit	5.0%	-1.5 pts	104%	110%	119%

EXHIBIT 3 The table drawn: five paths for the debt ratio



How to read: the horizontal axis counts years from today; the vertical axis is federal debt held by the public as a share of GDP, every line starting from today's 99% and running twenty years, with dots and a hairline at year ten, where the table's ten-year column and the black band are read; the end labels are year twenty. Every abundance line has the same 10% real growth and the same 3.49% interest rate on the debt. What differs between them is only the price path, falling 8% a year, held at 2% by the Fed, or a 5% nominal-income target, and the primary balance, the CBO's 2.6% deficit or a balanced budget. The gray line is today's economy with no robots, for comparison.

Same robots, same output, same real growth in every abundance row. The debt ratio goes to 152% or to 44% depending on whether prices fall or the central bank holds its target. That is the whole scenario in one variable, and it is a monetary variable. The abundance argument is a bet on Fed policy, which the academic form of the argument states openly and the popular form does not. Three things the table adds to that headline, and two things it cannot show.

The primary budget. The debt halves only in the row where the primary budget is balanced as well; at the CBO's current primary deficit, twelve percent nominal growth gets the ratio to 63% in a decade and does not halve it until year eighteen. At twenty years the two rows read 46% and 20%, and the deflation row 219%. A basic income sized at 8% of GDP on top of the current deficit holds the ratio flat at 100% even with twelve percent nominal growth, which is the full version of "abundance funds UBI": it funds it at the cost of the debt paydown, not in addition to it.

The coupon. The coupon cannot stay at 3.5% in either direction. If the average rate on the debt reprices toward nominal growth over five years, the Fed-holds-2% row dips to 93% and then climbs back to 110% by year ten; the paydown is a race between growth and the Treasury's term structure. The same repricing helps the deflation row, as it helped Japan, whose coupon fell to 1.4%: with the rate drifting down to match 1.2% nominal growth the ratio still rises, to 129% rather than 152%, because a 2.6% primary deficit accumulates whenever the interest rate is near the growth rate. At twenty years the repriced rows converge from opposite directions, the Fed-holds-2% row at 136% and the deflation row at 155%, which is the term-structure race run to its end: with the coupon following growth, the primary deficit decides.

The proponents' own row. The last row is the proponents' actual policy proposal, a nominal-income level target, at the 5% path its advocates typically cite. With 10% real growth that means prices falling 4.5% a year, and the ratio rises to 110% in a decade and 119% in two, and to 122% and 148% if the coupon reprices toward 5% as well, because a 5% nominal path sits below the coupon plus the primary deficit. A nominal target only pays down the debt if the level is set well above both.

What cannot be inflated away. The table is a table of dollar claims, and the rows that pay the debt down pay it in cheaper dollars. A large part of what the federal government owes is not a dollar claim. Medicare and the health programs have spent \$1.8 trillion so far this fiscal year, 29% of all federal outlays, more than

Social Security and more than interest on the debt, and what they owe is not a sum but a service: a hip replacement for a patient of 66, a knee for a patient of 68, the mean ages in the registry that tracks the surgeries, which logged 160,000 primary hip and 288,000 knee replacements in 2024 from the surgeons who report to it, a floor on the national count rather than a total. That promise is denominated in surgeons, operating rooms and implants, not in dollars. Nominal growth does not shrink it; only a fall in the cost of producing a surgery does. So the Fed-holds-2% row retires the debt that is a claim on dollars and leaves untouched the debt that is a claim on care, and a government that can inflate the first cannot inflate the second. The same is true of the deflation row: a promise of surgeries is unaffected by the price path in either direction, so the 219% no more contains the care debt than the 44% does. The in-kind share sits outside every row, orthogonal to the one variable the table varies.

Most of the budget is like this. Widen the lens and the point covers most of what Washington spends. Add Social Security, the income-security programs and veterans' benefits, which are paid in dollars indexed to prices, with veterans' medical care in kind, and the promises that are in kind or indexed come to about two-thirds of this fiscal year's outlays on the same Treasury table. The rows that pay the debt down act on the dollar claim, the Treasury's own nominal debt, which is the part of what the state owes that a price level can shrink; the inflation-indexed part, about \$2.2 trillion of \$31.5 trillion in marketable debt, does not shrink that way, and even the nominal part reprices as it rolls. The reverse holds too, and it is the proponents' strongest fiscal point: care is half labor, and if robots and AI reach the operating room the entitlement deflates in real terms with no policy decision at all. The nearest the table comes to showing that is the balanced-primary-budget rows, because a surgery that costs less is one of the ways that budget closes.

Fisher, and the lived case. Irving Fisher wrote the mechanism in 1933: "The more the debtors pay, the more they owe." By March of that year, liquidation had cut nominal debts 20% while the dollar's value rose 75%, so real debt had risen 40%. Japan is the modern instance and it is a productivity economy, not a failing one. From 1998 to 2012 Japanese consumer prices fell 0.2% a year on average, nominal GDP fell 0.6% a year while real GDP grew 0.6%, and the ten-year government bond averaged 1.4%. The interest rate exceeded nominal growth by about two points for fourteen years and general-government debt went from 102% of GDP in 1998 to 197% in 2012 and 200% by 2015. Japan survived it only because the coupon collapsed: interest payments in the fiscal 2026 budget are

¥13.1 trillion, against ¥11.2 trillion in 1990 on a debt many times smaller. Since 2013, with nominal growth at 2.3% and the ten-year at 0.4%, the spread has been minus two points; the ratio plateaued near 200%, spiked to 229% in the pandemic year of 2020, and has come down to 207% in 2025. The country did not grow its way out; it changed the sign of one spread.

152% or 44%

Debt held by the public as a share of GDP after ten years of the same 10% real growth: the first if prices are allowed to fall 8% a year, the second if the Federal Reserve holds 2% and the primary budget is balanced. Nothing about the robots differs between the two.

QUESTION FOUR

Who pays, and can a basic income be funded?

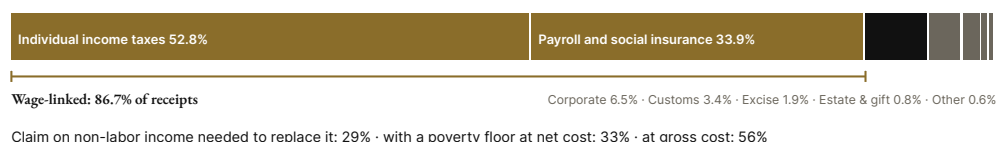
Question three showed that the debt ratio can fall under abundance if prices hold and the primary budget closes. The fourth question is whether it can close: the abundance argument's fiscal leg assumes a richer economy funds both the debt and a basic income, so the ledger has to ask what the federal government actually taxes today and what happens to that base if the robot-hours in question one arrive.

What it taxes. Fiscal year to date through July 2026, the Treasury has collected \$4.49 trillion. Individual income taxes are 52.8% of it and social insurance, mostly payroll taxes, another 33.9%. Corporate income taxes are 6.5% and customs duties 3.4%. On the national-accounts definition, which nets refunds and timing, personal current taxes are 43% of federal receipts and social insurance contributions 35%. Either way, between 78% and 87% of what Washington collects is tied to a wage. The CBO's fiscal 2025 actuals are the same picture in dollars: \$2.7 trillion from individuals, \$1.7 trillion from payroll, \$452 billion from corporations.

SOURCE OF FEDERAL RECEIPTS	FISCAL 2026 TO DATE	SHARE
Individual income taxes	\$2,369 bn	52.8%
Social insurance and retirement (payroll)	\$1,523 bn	33.9%
Corporate income taxes	\$293 bn	6.5%
Customs duties	\$154 bn	3.4%
Excise, estate and gift, other	\$147 bn	3.3%

EXHIBIT 4 The wage-linked base and the claim on capital

Federal receipts by source, fiscal 2026 to date (Monthly Treasury Statement)



How to read: the bar is every dollar the federal government has collected this fiscal year, divided by source from left to right; the two gold segments are the receipts tied to a wage, and the bracket beneath them measures their combined share. The last line states how large a claim on all non-labor income in the economy would be needed to replace those receipts, alone and with a basic income added at its net and its gross cost.

What happens to it. Korinek and Lockwood, in the first paper to treat this formally, put the leak at ten to fifteen cents of federal revenue for every dollar of value that moves from labor to capital, and note that labor's share has already fallen from 68% in 2000 to 58% by their measure. Their central result is a Laffer curve for labor taxes. A Laffer curve is the arc that revenue traces as a tax rate rises: nothing is collected at a rate of zero, nothing at a rate so high that the taxed activity stops, and the most somewhere between. Theirs says the peak of that arc for taxes on wages falls toward zero as the capital share of income goes to one, because there are fewer and fewer wages left to tax at any rate. Brookings' labor-shock scenario, in which the marginal share of new income going to labor falls from 0.54 to 0.30, halves the deficit improvement that its conventional scenario delivers, and the authors name the shift from highly taxed labor income to lightly taxed capital income as the reason. The CBO's own December 2024 review names both channels without sizing either: "different categories of income are taxed at different rates," and "if workers were left permanently unemployed or were reallocated to lower-paying jobs by the

technology, income and payroll taxes would decline." On question one's path the wage base does not erode at the margin; it goes.

The claim on capital. The identity is simple. Compensation of employees is \$16.2 trillion at an annual rate, half of gross domestic income. Non-labor income, everything else, is \$16.0 trillion. Federal receipts tied to wages are \$4.6 trillion. To replace them from capital income alone would take an effective claim of 29% on all non-labor income in the economy. That is before any basic income.

WHAT THE TRANSFER HAS TO COVER	CLAIM ON NON-LABOR INCOME
Replace today's wage-linked federal receipts	29%
Plus a poverty-line basic income at its net cost	33%
Plus that basic income at its gross cost	56%

The basic income line. The 2026 poverty guideline for one person is \$15,960. Paid to the 275 million Americans aged sixteen and over, that is \$4.4 trillion a year, 13.5% of GDP, which is the number the argument's critics cite. Widerquist's 2017 calculation is the number its designers cite: a poverty-level income with a 50% clawback as earnings rise costs 13% of its gross figure, because most of it is paid to people who pay it back in tax. Applied to today's guideline that is about \$570 billion, 1.8% of GDP, and about \$950 billion at a gentler 35% clawback. The difference between 13.5% and 1.8% of GDP is the difference between a transfer and a floor. Both ratios are to today's GDP; a floor fixed in real dollars shrinks as a share of a 10%-growth economy, to about 5.2% and 0.7% of GDP after a decade, which is the proponents' strongest point on affordability and does not change the source: neither is fundable from a wage base that has been automated away, and both require the capital claim above.

The answer to question four. The bill is paid from capital income or it is not paid, and every serious treatment ends in the same place. Altman's American Equity Fund would take 2.5% of the value of large companies each year, in shares, and 2.5% of the value of privately held land, and pay the proceeds to every adult. Korinek and Lockwood's sequence runs from consumption taxes while humans still consume, to a harvesting tax on autonomous capital at a rate set by how much society discounts the future, about 4% a year at a 4% discount rate, with sovereign wealth funds and public equity stakes as the instruments that capture AI returns without discouraging investment, and fixed-factor taxes raising 5% to 10% of GDP. Nayebi's threshold model finds that the capability the machines

need to fund a basic income falls by nearly half if public capture of AI rents rises from today's 15% to a third, and that concentrated markets lower the threshold because they raise the rents available to tax. Growth rate barely enters any of these results. Ownership share and capture rate are the whole answer. Which is why the capability-threshold version of the argument is arguing about the wrong variable.

Σ THE BALANCE

What the four answers sum to

Part 1 asked whether the robot exists; this note asks what follows if it does. The four answers, on the industry's own numbers, chain into one reading.

- **The robot cheapens what a factory makes.** Intelligence fell 78 to 114 times in price in under two years; goods have been deflating in relative terms since 2000; on the industry's path robot labor reaches a sixth of US hours inside the budget window even on the conservative one-shift case, and over half on the companies' own duty cycle; on the consensus path it is a twentieth by 2035.
- **It does not cheapen what a citizen has to buy.** Half the consumer basket is shelter, energy and medical care, and all three have risen faster than the index for twenty-five years while goods prices went nowhere. Land's share of a home's value has gone from a third to nearly two-thirds since 2012. A transfer paid into that market is a transfer to the fixed factor unless the fixed factor is taxed.
- **Whether the debt falls is a monetary decision, not a robotic one.** The abundance path pays down the debt only if productivity arrives as nominal income, only if the primary budget closes, and only until the Treasury's coupon catches up with growth; the part of what the state owes that is a promise of care rather than dollars, 29% of outlays, sits outside every row and falls only with the cost of care, half of which is labor. Falling prices with the same growth push the ratio past 150% in ten years and past 200% in twenty. Japan ran that case and needed fourteen years and a collapsing coupon to survive it.
- **The primary budget cannot close on a wage base that has gone.** Between 78% and 87% of federal revenue is wage-linked. Replacing it takes a 29%

claim on all non-labor income before a single dollar of basic income; a poverty floor adds four points, a full transfer adds twenty-seven, and a floor fixed in real dollars shrinks to under one percent of GDP after a decade of the growth the proponents forecast. The number that settles the argument is not the learning rate of a humanoid. It is the share of capital income the state is legislated to capture. The proposals exist, from George to Altman to Korinek and Lockwood; none has been enacted.

≡ **THOUGHT EXPERIMENT · NOT MEASURED**

The limit: what survives full abundance

This section is a thought experiment, and it is labeled as one. Nothing new in it is measured, and nothing in the four questions depends on it. It exists because the proponents' claim is a claim about the limit, and the ledger should say what survives there.

Assume the degree of abundance the claim requires: robots that build robots, factories, turbines, wires, transformers, and the survey and inspection work behind a permit, at a cost falling toward the energy and materials they contain. Every manufactured input deflates, including most of the grid this note treats as slow on its own horizon, and most of the labor in medical care. Three things do not.

The first is land, including the land under the wire. A transmission corridor is a right-of-way, a solar or wind farm at the scale the claim needs is acreage, and at the limit the energy bottleneck collapses into the land bottleneck rather than standing beside it. The second is permission. A robot can make a permit application cheap; it cannot make a county say yes. A veto is not an input and has no learning rate, but it has a legislature; the proponents' case at the limit is that the veto is repealed, which is a political act with no cost curve to cite either way. The third is physics, in Musk's own words at the same forum: "there will still be constraints on power, like electricity and mass."

Two things follow. The first favors the proponents. No missing machine stands between the present and their limit, only land and consent, which is a stronger position than the three-machines frame gives them, and it is why the second claim in the steelman, that the fixed factors are fixed by regulation rather than by nature, is the strongest form of their case. The second is the note's. If the only things that stay scarce at the limit are land and the

permission to use it, then whatever money survives is denominated in land and permission, and the price record already shows the surplus moving there: land's share of a home's value rose from 36% to 59% in twelve years. Medical care sits on the soft side of the line, half labor that AI reaches and half licensing that only a legislature can move.

What the thought experiment does not say is when. The four questions run ten to twenty years, and on that horizon energy binds by degree and land binds by regulation. The limit is a direction, not a date.

⊕ CODA · THE MONEYLESS ECONOMY

What Voyager knew: the other bottlenecks, and why the moneyless economy failed

Musk's prediction has two halves. The first, that robots and AI make labor abundant, is the half Part 1 and the first question of this note have measured. The second, that abundance ends money, is the half he leaves to the novels, and it is the half Star Trek's writers had to work out in detail, because a franchise without money has to explain to its audience, week after week, why nobody pays. Their answer was not robots. It was three machines: a replicator that turns energy into any object, a warp core that supplies that energy without a fuel bill, and a transporter that puts a person anywhere without owning the ground they stand on. Labor abundance was incidental. Land, energy and matter abundance were the premise.

Which of the three the real economy has. The replicator, in part: the price of a fixed unit of intelligence fell 78 to 114 times in under two years, and the goods that factories make have been deflating in relative terms since 2000. The transporter, not at all: land's share of a home's value rose from 36% to 59% between 2012 and 2024, and rent is up 148% since 2000, because a robot can frame a house in an hour and cannot make more of the places people want to live. The warp core, not at all: retail electricity is up 141% since 2000 while the solar module fell 99.7%, because the gap between the module and the meter is wires, capacity and regulation; the first two have learning rates the retail price has not yet shown on this window, and the third has none, and the data centers behind the robots are set to more than double their draw by 2030. Medical care sits between: half of it is labor that AI can plausibly reach, and the other half is licensing, liability and reimbursement. These are the other bottlenecks on the

ledger's horizon, and the thought experiment above says which of them survive the limit; they are the ones that decide whether a dollar is still needed, because an economy in which labor is free but land, energy and licensed care are not is an economy in which money is needed for exactly the things a citizen cannot do without.

Why it fails: the ration slips. The franchise ran the experiment. In a 1995 episode of Star Trek: Voyager, a starship stranded decades from home loses its energy surplus. The crew is already living on replicator rations, a fixed allowance of the energy the food machine consumes, and in the same episode the captain is dissuaded from spending one of hers on a cup of coffee. By later episodes the crew is trading the rations, staking them in a sweepstakes on the next day's particle count and paying a week's worth for a clarinet. The wiki that catalogues the series describes the outcome in one line: the crew "treated rations on replicator and holodeck use as currency." Energy became scarce; an energy-denominated money appeared by later episodes. The Federation's other money, latinum, exists for the mirror reason: by the production lore, it is the one substance the replicator cannot make, so it is the one thing worth trading. Money was not abolished by abundance. It was dormant, and it woke the moment one essential input became scarce, denominated in that input.

What that implies for the abundance claim. Read literally, "money will stop being relevant" is a claim that land, energy and licensed care will become as abundant as labor, and Musk states the constraint himself from the same podium, and keeps his conclusion: "There will still be constraints on power, like electricity and mass. The fundamental physics elements will still be constraints, but I think at some point currency becomes irrelevant." The note reads that sentence the other way. A constraint on an input people still compete for has to be rationed, and a ration that can be traded is a price; the Voyager writers reached the same result in five episodes. So the endless-abundance claim reduces to this note's second question, and the price record since 2000 answers it in the wrong direction for the claim. The proponents do not dispute which inputs bind; Musk's own version of the argument is energy-first, and the module and cell cost curves in the steelman are real. The disagreement is whether those curves reach the meter and the lot line, and the record so far says they have not. That is not a finding that it can never be true. The reversal block states what would change the reading: real rent and real electricity falling for four consecutive quarters, with land's share of home value turning down. Until then, the claim that robots end money is a claim about energy and land policy, not

about robots, and the people best placed to make it true are utility regulators and zoning boards rather than roboticists. Abolish money and it comes back the moment a fixed factor binds, and it comes back denominated in whichever factor bound.

≡ WHAT THIS LEDGER CANNOT SEE

Risk factors

- **The hours table in question one is an upper bound built on the highest forecast.** Bank of America's shipments are worldwide and the US share is unknown; the table treats all of them as domestic, and the bank's 2035 volume is four times the consensus path Part 1 uses for the house's own case. The duty cycles and the five-year life are the companies' assumptions from Part 1, which no company has published as an outcome. The path is used as the proponents' own case, not as a house forecast; on the house path every date in this note moves four to five years later, the later ones by extrapolation beyond the modeled window.
- **The debt table holds the coupon fixed in every row.** That flatters the high-growth rows and penalizes the deflation rows equally: Japan's coupon fell to 1.4% during its deflation, and a Treasury borrowing into 8% price declines would not pay 3.49%. The repricing sensitivity is run both ways: the Fed-2% row turns back up to 110% by year ten, and the deflation row rises to 129% rather than 152% with the coupon drifting to 1.2%, or 131% floored at Japan's 1.4%. The truth is between fixed and fully repriced, set by the Treasury's maturity profile, which the note does not model. The twenty-year column compounds a fixed coupon for two decades and is directional only; with the coupon repriced, the Fed-holds-2% row ends at 136% and the deflation row at 155%.
- **The primary balance is the CBO's 2026 figure held constant.** The CBO itself has it improving to 2.1% by 2036. A 10% real growth economy would change every line of the budget in ways no baseline captures; the rows are arithmetic on stated inputs, not projections.
- **The adult count is the civilian population aged sixteen and over.** An eighteen-and-over base would be smaller by several percent. Widerquist's ratio comes from a 2015 design with a \$12,000 benefit and is applied here as a proportion only; a differently designed floor would net differently.

- **"Wage-linked" is an upper bound on the wage base.** Individual income tax receipts include capital gains and other non-wage income, so some of the 87% already comes from capital.
- **Japan's debt series exist in two IMF vintages** that differ by up to thirty points in some years. The note uses the current DataMapper vintage; the older FRED mirror gives higher levels and the same shape.
- **The Star Trek closing section is words only** and relies on a fan-maintained encyclopedia for quotations; the point that latinum cannot be replicated is a production-source inference rather than an on-screen line.

↪ REVERSAL

What would change our read

1. **Nominal GDP growth running more than four points above the average interest rate on the debt for eight consecutive quarters while the primary deficit shrinks.** That is the spread the abundance case needs, and it is published monthly by the Treasury and quarterly by the BEA. If it prints, the debt row is moving the proponents' way regardless of the Fed's stated framework.
2. **The Federal Reserve adopting a nominal-income or price-level target, in its Statement on Longer-Run Goals or its operating practice.** The January 2026 statement reaffirms a 2% inflation rate with no make-up language. A change in that document changes the decisive variable in question three.
3. **Real rent and real electricity falling for four consecutive quarters, with the AEI land share of home value turning down.** The fixed-factor reading fails if the fixed factors start to deflate; the series are monthly and annual respectively. This condition also governs the thought experiment: if the meter and the lot line begin to follow the module inside the ledger's window, the limit is arriving as a date, and the second question is answered the proponents' way.
4. **Enacted legislation establishing a claim on capital or land income of the order question four requires,** a sovereign wealth fund, an equity levy, a land-value tax or a capital-harvesting tax with a stated rate. The note's central objection is that no such claim has been enacted; one would remove it.

5. **Part 1's robot-hour path failing to materialize**, meaning audited deliveries below a tenth of Bank of America's forecast by 2028. The hours table in question one is the input to everything downstream; if the hours do not come, the fiscal question moves out of the budget window and this note's urgency goes with it, in the proponents' disfavor on capability but in their favor on the transition.
6. **Federal receipts from capital income rising as a share of the total for three consecutive fiscal years without a change in law**. That would mean the base was migrating on its own, and the 29% claim was being collected through existing instruments.
7. **The average interest rate on the debt exceeding nominal GDP growth for eight consecutive quarters while the primary deficit does not shrink**. That is the deflation row's signature arriving in the data, and it would confirm the note's adverse case as firmly as condition 1 would confirm the proponents'.

Tripwires, re-scored on release: Treasury average interest rates (monthly) and debt to the penny (daily); BEA GDP (quarterly); Monthly Treasury Statement receipts by source; CPI shelter, electricity and medical (monthly); AEI land-price indicators (annual); Fed Z.1 (quarterly); FOMC framework statement (annually in January); CBO Budget and Economic Outlook; Part 1's tripwires for the robot path.

APPENDIX

Sources & method

Debt arithmetic: debt held by the public as a share of GDP from FRED (FYGFGDQ188S, 2026 Q1); average interest rate on total interest-bearing debt from Treasury Fiscal Data (avg_interest_rates, August 2026); nominal and real GDP (BEA via FRED, 2026 Q2); federal interest payments (A091RC1Q027SBEA); primary deficit and projections from the CBO Budget and Economic Outlook 2026–2036. Debt-ratio recursion over a twenty-year horizon (reported at years 5, 10 and 20), scenarios and the coupon-repricing sensitivity in `data/debt_path.py`. Japan: IMF World Economic Outlook DataMapper (gross debt, CPI, real growth, fiscal balance; archived JSON), FRED for nominal and real GDP and the ten-year JGB yield, Japan Ministry of Finance Public Finance Fact Sheet April 2026; averages in `data/japan.py`. Fisher, "The Debt-Deflation Theory of Great Depressions," *Econometrica* 1933.

Robot-hour path: Part 1's `wrights_law.json` (Bank of America shipment path and cumulative units) and `robot_hour_model.json` (duty cycles, success rates, service life); US hours from BLS payrolls and average weekly hours (FRED PAYEMS, AWHAEPT, August 2026); `data/bridge_robot_hours.py`.

Prices and fixed factors: CPI components (BLS via FRED, July 2026, seasonally adjusted) indexed to January 2000; BLS relative importance of CPI components, December 2025; AEI Housing Center land

price and land share indicators 2012–2024 (national CSV archived); Larson, BEA WP2015-3 (US land value 2009); Fed Financial Accounts Z.1 table B.101 via FRED (2026 Q1); CMS National Health Expenditure Accounts (2024 data, 2034 projection); IEA, Energy and AI (April 2025), report PDF pp. 14, 63–64 (archived); UBC Physics C21 on human metabolic power; Ricardo, Principles (1817) ch. 2; George, Progress and Poverty (1879); Aghion, Jones and Jones, NBER 23928. Computation in `data/what_deflated.py`.

Tax base and transfers: Monthly Treasury Statement Table 4, fiscal year to date through July 2026, via Treasury Fiscal Data; NIPA federal receipts by type (FRED FGRECPT, A074RC1Q027SBEA, W780RC1Q027SBEA, B075RC1Q027SBEA); compensation of employees and gross domestic income (FRED COE, GDI); HHS 2026 poverty guidelines; BLS civilian noninstitutional population 16+ (FRED CNP16OV); Widerquist, "The Cost of Basic Income," Basic Income Studies 2017; Korinek and Lockwood, "Public Finance in the Age of AI: A Primer," NBER w34873 (Feb 2026) and "Preserving Fiscal Stability in the Age of Transformative AI," Digitalist Papers; Nayebi, arXiv 2505.18687 (v1 May 2025, v4 July 2026); Altman, "Moore's Law for Everything" (March 2021); Brookings, "Can AI restore fiscal sustainability in the US?" (July 2026); Yale Budget Lab, "What Might AI Adoption Mean for the Fiscal and Economic Outlook?" (updated May 2026); Penn Wharton Budget Model, generative AI and productivity (2025); CBO, "Artificial Intelligence and Its Potential Effects on the Economy and the Federal Budget" (Dec 2024). Computation in `data/tax_base.py`, inputs in `data/ubi_inputs.json`.

Monetary framework: FOMC Statement on Longer-Run Goals and Monetary Policy Strategy, reaffirmed January 27, 2026; Mercatus Center notes on nominal GDP level targeting (Fuhrig 2019, Koenig 2024).

Entitlements in kind: Monthly Treasury Statement Table 9, outlays by function, fiscal 2026 to date through July 2026 (Treasury Fiscal Data `mts_table_9`, archived); AAOS American Joint Replacement Registry 2025 Annual Report, Tables 2.1, 2.2, 3.1 and 3.2 (PDF and text archived; registry capture is voluntary and incomplete, so counts are a floor). Notes in `data/sources/entitlements/`.

The proponents' claim and horizon: Musk at the U.S.-Saudi Investment Forum, Washington, November 19, 2025 (C-SPAN recording; transcript by The Singju Post, archived) and at Viva Technology, May 2024 (via Fortune, January 19, 2026).

Closing section: Memory Alpha entries for "Replicator ration," "The Cloud," "Money" and "Latinum" (wikitext archived).

Limitations: the debt table is arithmetic on stated inputs with a fixed coupon and a fixed primary balance, not a projection; the robot-hour path is the industry's forecast and an upper bound; the transfer costs depend on design and are shown as a range; every conclusion is dated to the vintages above and re-scores on the tripwires listed.

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